

%L0: Problem 2

```
n1=[1 1];  
d1=[1 2];  
n2=[1];  
d2=[500 0 0];  
g1=tf(n1,d1)
```

g1 =

$$\frac{s + 1}{s + 2}$$

Continuous-time transfer function.
Model Properties

```
g2=tf(n2,d2)
```

g2 =

$$\frac{1}{500 s^2}$$

Continuous-time transfer function.
Model Properties

%a. Cascade control g1.g2

```
ns=conv(n1,n2);  
ds=conv(d1,d2);  
gc=tf(ns,ds)
```

gc =

$$\frac{s + 1}{500 s^3 + 1000 s^2}$$

Continuous-time transfer function.
Model Properties

```
[n,d]=series(n1,d1,n2,d2);  
printsys(n,d)
```

num/den =

$$\frac{s + 1}{500 s^3 + 1000 s^2}$$

%-----

%b. Parallel control g1+g2

```
gp=g1+g2
```

gp =

$$\frac{500 s^3 + 500 s^2 + s + 2}{500 s^3 + 1000 s^2}$$

Continuous-time transfer function.
Model Properties

```
[np,dp]=parallel(n1,d1,n2,d2)
```

```
np = 1x4
    500    500    1    2
dp = 1x4
    500    1000    0    0
```

```
printsys(np,dp)
```

num/den =

$$\frac{500 s^3 + 500 s^2 + s + 2}{500 s^3 + 1000 s^2}$$

```
%-----
```

```
%c.1. Feedback control : unity +ve feedback
disp("unity positive")
```

unity positive

```
[nf1,df1]=feedback(ns,ds,1,1,-1)
```

```
nf1 = 1x4
    0    0    1    1
df1 = 1x4
    500    1000    1    1
```

```
printsys(nf1,df1)
```

num/den =

$$\frac{s + 1}{500 s^3 + 1000 s^2 + s + 1}$$

```
%c.2. Feedback control : unity -ve feedback
disp("unity negative")
```

unity negative

```
[nf2,df2]=feedback(ns,ds,1,1,1)
```

```
nf2 = 1x4
    0    0    1    1
df2 = 1x4
    500    1000    -1    -1
```

```
printsys(nf2,df2)
```

num/den =

$$\frac{s + 1}{500 s^3 + 1000 s^2 - 1 s - 1}$$

%c.3. g1 feedback, g2 forward feedback

```
[nf3,df3]=feedback(n2,d2,n1,d1,-1)
```

```
nf3 = 1x4
```

```
0 0 1 2
```

```
df3 = 1x4
```

```
500 1000 1 1
```

```
printsys(nf3,df3)
```

num/den =

$$\frac{s + 2}{500 s^3 + 1000 s^2 + s + 1}$$

%L0: Exercise 1

```
n4=[6 0 1];
```

```
d4=[1 3 3 1];
```

```
g=tf(n1,d1)
```

g =

```
s + 1
```

```
-----
```

```
s + 2
```

Continuous-time transfer function.

Model Properties

```
s=tf('s');
```

```
h=((s+1)*(s+2))/((s+2i)*(s-2i)*(s+3))
```

h =

```
s^2 + 3 s + 2
```

```
-----
```

```
s^3 + 3 s^2 + 4 s + 12
```

Continuous-time transfer function.

Model Properties

%i. roots of G(s)

```
z1=roots(n4)
```

```
z1 = 2×1 complex  
0.0000 + 0.4082i  
0.0000 - 0.4082i
```

```
p1=roots(d4)
```

```
p1 = 3×1 complex  
-1.0000 + 0.0000i  
-1.0000 - 0.0000i  
-1.0000 + 0.0000i
```

```
%H(s) as ratio of two polynomials
```

```
hn1=[1 1];  
hn2=[1 2];  
hd1=[1 2*i];  
hd2=[1 -2*i];  
hd3=[1 3];  
numh=conv(hn1,hn2);  
denh=conv(conv(hd1,hd2),hd3);  
H=tf(numh, denh)
```

```
H =
```

$$\frac{s^2 + 3s + 2}{s^3 + 3s^2 + 4s + 12}$$

```
Continuous-time transfer function.  
Model Properties
```

```
%iii. g(s)/h(s) and pzmap
```

```
numgh=conv(n4,denh);  
dengh=conv(d4,numh)
```

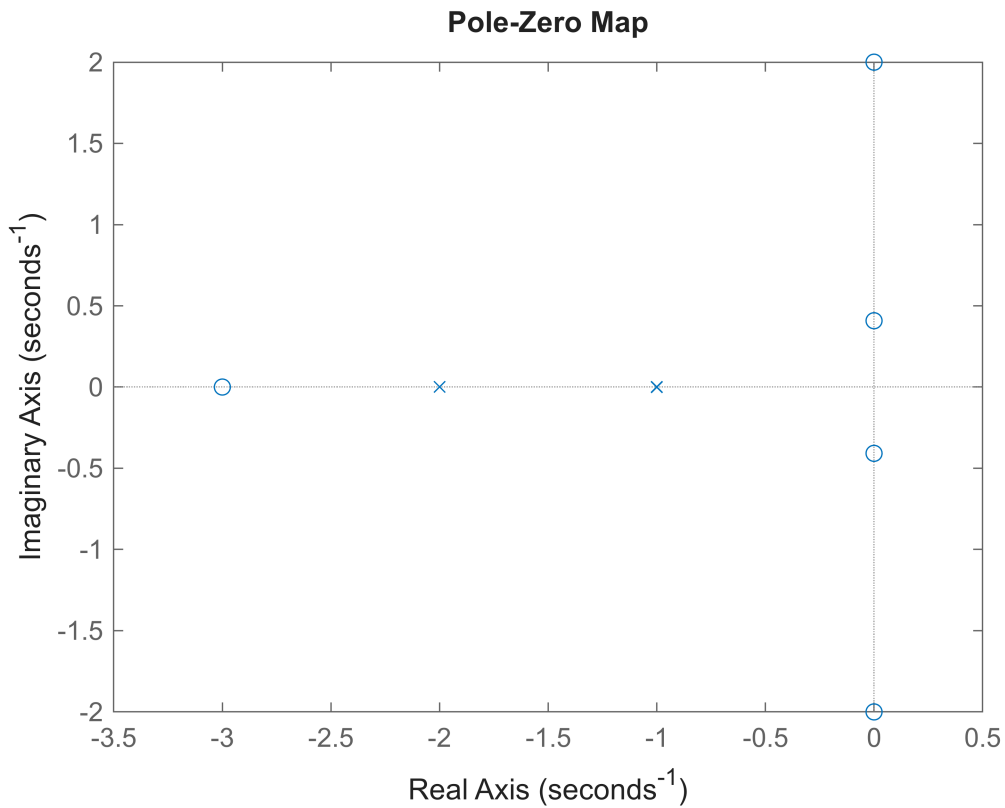
```
dengh = 1×6  
1 6 14 16 9 2
```

```
printsys(numgh, dengh)
```

```
num/den =
```

$$\frac{6s^5 + 18s^4 + 25s^3 + 75s^2 + 4s + 12}{s^5 + 6s^4 + 14s^3 + 16s^2 + 9s + 2}$$

```
pzmap(numgh,dengh)
```



%L0: Exercise 2

```
nps=[1 2 1];
qps=[1 1];
%i. p(s)q(s)
pq=tf(conv(nps,qps),1)
```

pq =

$s^3 + 3s^2 + 3s + 1$

Continuous-time transfer function.
Model Properties

```
%ii. poles and zeros of G(s) = q(s)/p(s)
[z,p,k]=tf2zp(qps,nps)
```

```
z =
-1
p = 2x1
-1
-1
k =
1
```

```
%iii. p(-1)
```

```
gpq=tf(qps,nps)
```

```
gpq =
```

$$\frac{s + 1}{s^2 + 2s + 1}$$

Continuous-time transfer function.
Model Properties

```
polyval(p,-1)
```

```
ans =  
0
```

%L0: Exercise 3

```
ne3=[1 6 8];  
de3=[1 8 12 16 20];  
gse3=tf(ne3,de3)
```

```
gse3 =
```

$$\frac{s^2 + 6s + 8}{s^4 + 8s^3 + 12s^2 + 16s + 20}$$

Continuous-time transfer function.
Model Properties

```
zpk(gse3)
```

```
ans =
```

$$\frac{(s+4)(s+2)}{(s+6.449)(s+1.551)(s^2 + 2)}$$

Continuous-time zero/pole/gain model.
Model Properties

```
pzmap(gse3)
```

